

# Extension of a homeomorphism

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We wish to prove the following theorem concerning metric spaces  $E$ ,  $F$ ,  $\overline{F}$ ,  $\overline{E}$ .

**I.** *If  $F$  is closed in  $E$ , then a homeomorphism between  $F$  and  $\overline{F}$  can be extended to a homeomorphism between  $E$  and a suitable space  $\overline{E}$ .*

In other words, the possibility of “re metrization” is asserted. Instead of the old distances  $uv$  defining the metric space  $E$ , we can introduce new distances  $\overline{uv}$  defining a space  $\overline{E}$  homeomorphic to  $E$  and taking the values prescribed by the metric of  $\overline{F}$  for pairs of points in  $F$ . (The homeomorphism means that  $uv \rightarrow 0$  with  $u$  fixed implies  $\overline{uv} \rightarrow 0$ , and conversely.) We can also say that a metrizable topological space  $E$  can always be endowed with a metric whose restriction to a closed set  $F \subset E$  is prescribed.

The proof of I is modelled on my earlier proof (Math. Zeitschrift 5 (1919), p. 296) of H. Tietze’s theorem that a continuous function on a closed set can be extended to a continuous function on the whole space, but requires considerably more devices.

We denote by  $u, v, w$  points of the space  $E$ , by  $a, b, c, p, q, r$  points of the closed set  $F$ , and by  $x, y$  points of the open complement  $G = E - F$ . Let

$$\delta(u) = \inf_p pu$$

be the infimum of the (old) distances of the point  $u$  from the points  $p \in F$  (the lower distance between  $u$  and  $F$ ), so that  $\delta(a) = 0$ ,  $\delta(x) > 0$ . Taking the infimum over  $p$  on both sides of  $pu \leq pv + uv$ , we obtain

$$|\delta(u) - \delta(v)| \leq uv.$$

Since we shall need it later, we consider at once the function

$$(0) \quad \delta(u, v) = \min[uv, \delta(u) + \delta(v)],$$

which is symmetric and vanishes if and only if  $u = v$  or  $u, v$  are two points of  $F$ . It satisfies the “triangle inequality”

$$\delta(u, v) + \delta(v, w) \geq \delta(u, w).$$

Indeed, the four inequalities

$$\begin{aligned} uv + uw &\geq uw \\ uv + [\delta(v) + \delta(w)] &\geq \delta(u) + \delta(w) \\ [\delta(u) + \delta(v)] + vw &\geq \delta(u) + \delta(w) \\ [\delta(u) + \delta(v)] + [\delta(v) + \delta(w)] &\geq \delta(u) + \delta(w) \end{aligned}$$

hold, and taking the least of the four expressions on each side proves the assertion.

Theorem I is equivalent to the following theorem.

**II.** *If  $F$  is closed in  $E$  and homeomorphic to  $\overline{F}$ , then there is a function  $\varphi(u, a) \geq 0$  with the following properties.*

- ( $\alpha$ )  $\varphi(b, a) = \overline{ab}$
- ( $\beta$ ) *If  $\delta(x) \rightarrow 0$ ,  $\varphi(x, a) \rightarrow 0$ , then  $ax \rightarrow 0$  (with  $a$  fixed).*
- ( $\gamma$ ) *For every pair of points  $u, v$ , the supremum*

$$(1) \quad \psi(u, v) = \sup_a |\varphi(u, a) - \varphi(v, a)|$$

*is finite and converges to 0 as  $uv \rightarrow 0$  (with  $u$  fixed).*

Indeed, if I holds, the new distance  $\varphi(u, a) = \overline{au}$  satisfies the requirements of II, with  $\psi(u, v) \leq \overline{uv}$ . Property ( $\beta$ ) holds because  $\overline{ax} \rightarrow 0$  implies  $ax \rightarrow 0$ , and ( $\gamma$ ) holds because  $uv \rightarrow 0$  implies  $\overline{uv} \rightarrow 0$ .

Conversely, to deduce I from II, we first observe that  $\psi(u, v)$  is symmetric and satisfies the triangle inequality

$$\psi(u, v) + \psi(v, w) \geq \psi(u, w)$$

This function cannot yet be used as the new distance, if only because it may vanish for  $u \neq v$ . We therefore combine it with the function (0) to form

$$(2) \quad \overline{uv} = \max[\psi(u, v), \delta(u, v)]$$

and show that this is a suitable distance. Certainly  $\overline{uv}$  is symmetric and satisfies the triangle inequality  $\overline{uv} + \overline{vw} \geq \overline{uw}$ .

Let us state property ( $\gamma$ ) separately for the three cases in which  $u, v$  belong respectively to  $F$ ,  $F$  or  $F, G$  or  $G, G$ .

$$(\gamma_1) \text{ We have } \psi(b, c) = \sup_a |\overline{ab} - \overline{ac}| = \overline{bc}.$$

$$(\gamma_2) \text{ The quantity } \psi(x, b) = \psi(b, x) = \sup_a |\varphi(x, a) - \overline{ab}| \geq \varphi(x, b) \text{ is finite and converges to 0 as } xy \rightarrow 0 \text{ (with } b \text{ fixed).}^1$$

$$(\gamma_3) \text{ The quantity } \psi(x, y) = \sup_a |\varphi(x, a) - \varphi(y, a)| \text{ is finite and converges to 0 as } xy \rightarrow 0 \text{ (with } x \text{ fixed).}$$

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<sup>1</sup> $xb \rightarrow 0$  with  $x$  fixed does not arise.

We now see that  $\overline{uv}$  vanishes when  $u = v$  and only then. Indeed,  $\overline{uv} = 0$  means  $\delta(u, v) = \psi(u, v) = 0$ . Thus either  $u = v$  or  $u = b, v = c$  are points of  $F$ , but even in the latter case  $\psi(b, c) = \overline{bc} = 0$ , so that  $b = c$ . Consequently,  $\overline{uv}$  defines a metric space  $\overline{E}$ , and it remains to prove that this space is homeomorphic to  $E$ .

If  $uv \rightarrow 0$  (with  $u$  fixed), then  $\delta(u, v) \rightarrow 0$  and, by  $(\gamma)$ ,  $\psi(u, v) \rightarrow 0$ , so that  $\overline{uv} \rightarrow 0$ .

Let  $\overline{uv} \rightarrow 0$  (with  $u$  fixed), so that  $\delta(u, v) \rightarrow 0$  and  $\psi(u, v) \rightarrow 0$ . If  $u = x$  is a point of  $G$ , then  $\delta(x, v) = \min[xv, \delta(x) + \delta(v)]$ , and since  $\delta(x) > 0$  is fixed,  $\delta(x, v) \rightarrow 0$  alone implies  $xv \rightarrow 0$ . If  $u = b$  is a point of  $F$ , then  $\delta(b, v) = \min[bv, \delta(v)] = \delta(v) \rightarrow 0$  and  $\psi(b, v) \rightarrow 0$ . Here we must distinguish the cases  $v = c \in F$  and  $v = y \in G$  (at least one of which must occur infinitely often as  $v$  varies). In the first case  $\psi(b, c) = \overline{bc} \rightarrow 0$  and  $bc \rightarrow 0$ . In the second case  $\delta(y) \rightarrow 0$  and  $\psi(b, y) \rightarrow 0$ , and hence also  $\varphi(y, b) \rightarrow 0$ , so that  $(\beta)$  implies  $by \rightarrow 0$ . Thus  $\overline{uv} \rightarrow 0$  always implies  $uv \rightarrow 0$ .

This proves that I follows from II. It remains to construct a function  $\varphi(u, a)$  with properties  $(\alpha)$   $(\beta)$   $(\gamma)$ , or a function  $\varphi(x, a)$  with properties  $(\beta)$   $(\gamma_2)$   $(\gamma_3)$ .

We form (with  $p \in F$ , as agreed) the infimum

$$(3) \quad \omega(x, c) = \inf_p \left[ \overline{cp} + 2 \frac{px}{\delta(x)} - 2 \right]$$

and the supremum

$$(4) \quad \varphi(x, a, c) = \sup_p \left[ \overline{ap} - \overline{cp} - \frac{px}{\delta(x)} + 1 \right],$$

which is certainly finite (even when  $\overline{F}$  is unbounded), since the expression in brackets is  $\leq \overline{ac}$ . Then

$$-\varphi(x, a, c) = \inf_p \left[ \overline{cp} - \overline{ap} + \frac{px}{\delta(x)} - 1 \right] \leq \omega(x, c)$$

and accordingly we put

$$(5) \quad \varphi(x, a) = \varphi(x, a, c) + \omega(x, c) \geq 0,$$

where  $c$  is any fixed point of  $F$ . The following considerations prove that  $\varphi(x, a)$  has the required properties.

(A) We choose a point  $p$  depending on  $x$  such that  $px < \delta(x) + \delta(x)^2$ , so that  $\frac{px}{\delta(x)} - 1 < \delta(x)$ . Then

$$\begin{aligned} \omega(x, c) &< \overline{cp} + 2\delta(x) \\ \varphi(x, a, c) &> \overline{ap} - \overline{cp} - \delta(x). \end{aligned}$$

If now  $bx \rightarrow 0$  with  $b$  fixed, then  $\delta(x) \rightarrow 0$ ,  $px \rightarrow 0$ ,  $bp \rightarrow 0$ , and hence  $\overline{bp} \rightarrow 0$ ,  $\overline{ap} \rightarrow \overline{ab}$ ,  $\overline{cp} \rightarrow \overline{cb}$ . We therefore obtain, as

$bx \rightarrow 0$ ,

$$\overline{\lim} \omega(x, c) \leq \overline{cb}, \quad \underline{\lim} \varphi(x, a, c) \geq \overline{ab} - \overline{cb}.$$

(B) We choose a point  $q$  depending on  $x$  (and  $c$ ) such that

$$(6) \quad \omega(x, c) + \delta(x) > \overline{cq} + 2 \frac{qx}{\delta(x)} - 2.$$

On the other hand,

$$\omega(x, c) \leq \overline{cb} + 2 \frac{bx}{\delta(x)} - 2.$$

Combining these inequalities, we obtain

$$2 \frac{qx}{\delta(x)} < \overline{cb} + 2 \frac{bx}{\delta(x)} + \delta(x)$$

$$2qx < 2bx + \overline{cb} \delta(x) + \delta(x)^2$$

It follows again that if  $bx \rightarrow 0$  (with  $b$  fixed), then  $qx \rightarrow 0$ ,  $bq \rightarrow 0$ ,  $\overline{bq} \rightarrow 0$ ,  $\overline{cq} \rightarrow \overline{cb}$ . From (6), or  $\omega(x, c) > \overline{cq} - \delta(x)$ , we obtain

$$\underline{\lim} \omega(x, c) \geq \overline{cb},$$

Together with (A), this shows that  $\omega(x, c) \rightarrow \overline{cb}$  as  $bx \rightarrow 0$ .

Furthermore, we also have

$$\varphi(x, a, c) \geq aq - \overline{cq} - \frac{qx}{\delta(x)} + 1,$$

and adding this to (6) yields

$$\varphi(x, a) + \delta(x) > \overline{aq} + \frac{qx}{\delta(x)} - 1$$

This contains the proof of  $(\beta)$ . Indeed, if  $\varphi(x, a) \rightarrow 0$  and  $\delta(x) \rightarrow 0$  with  $a$  fixed, then  $\overline{aq} \rightarrow 0$  and  $\frac{qx}{\delta(x)} - 1 \rightarrow 0$ , so that  $aq \rightarrow 0$ ,  $qx \rightarrow 0$ ,  $ax \rightarrow 0$ .

Finally, from  $\varphi(x, a) > \overline{aq} - \delta(x)$  we deduce

$$\overline{ab} - \varphi(x, a) < \overline{ab} - \overline{aq} + \delta(x) \leq \overline{bq} + \delta(x)$$

This proves one half of  $(\gamma_2)$ , because the positive bound  $\overline{bq} + \delta(x)$ , below which  $\overline{ab} - \varphi(x, a)$  remains, is independent of  $a$  ( $q$  depended only on  $x$  and on  $c$ , fixed once and for all). It is therefore a function of  $x, b$  alone and tends to 0 with  $bx$ , since we found above that  $\overline{bq} \rightarrow 0$ .

(C) We choose a point  $r$  depending on  $x, a$  (and  $c$ ) such that

$$(7) \quad \begin{aligned} \varphi(x, a, c) - \delta(x) &< \overline{ar} - \overline{cr} - \frac{rx}{\delta(x)} + 1 \\ &\leq \overline{ac} - \frac{rx}{\delta(x)} + 1; \end{aligned}$$

On the other hand,

$$\begin{aligned}\varphi(x, a, c) &\geq \overline{ab} - \overline{cb} - \frac{bx}{\delta(x)} + 1 \\ &\geq -\overline{ac} - \frac{bx}{\delta(x)} + 1,\end{aligned}$$

and combining the two yields

$$\begin{aligned}\frac{rx}{\delta(x)} &< 2\overline{ac} + \frac{bx}{\delta(x)} + \delta(x), \\ rx &< bx + 2\overline{ac} \cdot \delta(x) + \delta(x)^2.\end{aligned}$$

Again it follows that if  $bx \rightarrow 0$  with  $b$  fixed, then also  $rx \rightarrow 0$ ,  $br \rightarrow 0$ ,  $\overline{br} \rightarrow 0$ ,  $\overline{ar} \rightarrow \overline{ab}$ ,  $\overline{cr} \rightarrow \overline{cb}$ . Thus (7), or  $\varphi(x, a, c) < \overline{ar} - \overline{cr} + \delta(x)$ , implies  $\lim \varphi(x, a, c) \leq \overline{ab} - \overline{cb}$ . Together with (A), this shows that  $\varphi(x, a, c) \rightarrow \overline{ab} - \overline{cb}$  as  $bx \rightarrow 0$ .

Adding the limit relations

$$\omega(x, c) \rightarrow \overline{cb}, \quad \varphi(x, a, c) \rightarrow \overline{ab} - \overline{cb}$$

which hold as  $bx \rightarrow 0$ , we obtain  $\varphi(x, a) \rightarrow \overline{ab}$ , and in particular  $\varphi(x, b) \rightarrow 0$ .

(D) Adding

$$-\overline{cp} - \frac{px}{\delta(x)} + 1$$

to both sides of  $\overline{ap} \leq \overline{bp} + \overline{ab}$  and taking the supremum over  $p$ , we obtain

$$\varphi(x, a, c) \leq \varphi(x, b, c) + \overline{ab},$$

and then, by adding  $\omega(x, c)$ ,

$$\varphi(x, a) \leq \varphi(x, b) + \overline{ab}.$$

This contains the second half of  $(\gamma_2)$ . For, just as  $\overline{ab} - \varphi(x, a)$  does, the quantity  $\varphi(x, a) - \overline{ab}$  now has an upper bound  $\geq 0$  which is independent of  $a$  and depends only on  $x, b$ , namely  $\varphi(x, b)$ , and this tends to 0 as  $bx \rightarrow 0$ .

(E) To prove  $(\gamma_3)$ , we observe that

$$px \cdot \delta(y) - py \cdot \delta(x) = [px - py]\delta(y) + py[\delta(y) - \delta(x)]$$

has absolute value  $\leq xy \cdot [\delta(y) + py] \leq 2xy \cdot py$ . Without loss of generality, we may take the space  $E$  to be bounded, say of diameter  $\leq \frac{1}{2}$ , since every space is homeomorphic to a bounded one (replace the distances  $\delta$  by  $\frac{\delta}{1+\delta}$ , which preserves the triangle inequality. By contrast, assuming that  $\overline{F}$  is bounded would restrict the scope of Theorem I). We therefore have

$$\left| \frac{px}{\delta(x)} - \frac{py}{\delta(y)} \right| \leq \frac{xy}{\delta(x)\delta(y)}$$

and, by the same reasoning as in (D),

$$\begin{aligned} |\omega(x, c) - \omega(y, c)| &\leq \frac{2xy}{\delta(x)\delta(y)}, \\ |\varphi(x, a, c) - \varphi(y, a, c)| &\leq \frac{xy}{\delta(x)\delta(y)}, \\ |\varphi(x, a) - \varphi(y, a)| &\leq \frac{3xy}{\delta(x)\delta(y)}, \end{aligned}$$

The right-hand side is a bound independent of  $a$  which tends to 0 as  $xy \rightarrow 0$  with  $x$  fixed (when  $\delta(x) \rightarrow \delta(y) > 0$ ).

This completes the proof of Theorem I.

Two simple consequences are the following theorems, the second of which was proved by a different method by V. Niemytzki and A. Tychonoff (Fund. Math. 12 (1928), pp. 118–120).

*A metrizable space is compact if and only if it is bounded in every metric.*

*A metrizable space is compact if and only if it is complete in every metric.*

The necessity of the conditions is clear. Conversely, if the space  $E$  is not compact (compact in itself), then it contains a countable set  $F = \{a_1, a_2, \dots, a_n, \dots\}$  without an accumulation point, which is therefore closed in  $E$ . Since two discrete sets of the same cardinality are always homeomorphic,  $F$  is homeomorphic both to the set of numbers  $F_1 = \{1, 2, \dots, n, \dots\}$  and to

$F_2 = \left\{1, \frac{1}{2}, \dots, \frac{1}{n}, \dots\right\}$ . By I,  $E$  is therefore homeomorphic to an

unbounded space  $E_1$  and to an incomplete space  $E_2$  (in which  $F_2$  is closed, so that there is a Cauchy sequence without a limit).